

TECHNICAL VALIDATION REPORT

Full-Upright MBB Beam

True 3D FEM screening of cubic, gyroid, and honeycomb infill at 15% and 25%

Document	Full-upright six-case FDM infill FEM comparison
Prepared	23 July 2026
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Executive Summary

This report compares three linear-elastic finite-element cases using one complete upright topology-optimized MBB beam. Geometry, loading, supports, material baseline, and mesh resolution are held constant so that infill pattern and density are the comparison variables.

KEY RESULT At 100 N, gyroid 25% is the stiffest sparse screening case at 314.92 N/mm. Its mean load-point displacement is 2.7% lower than cubic 25%.

The sparse cases are validated as comparative numerical screening models. They are not yet validated as printed-part strength or failure predictions because each sparse core is homogenized and requires coupon or representative-volume calibration.

1. Purpose and Scope

The purpose is to provide a reproducible comparison of cubic, gyroid, and honeycomb infill for the recovered MBB beam while eliminating orientation and geometry differences between slicer archives. All accepted cases use the same full upright STL. The supplied 25% half-beam archive is used only to verify their density-specific toolpaths.

2. Model Definition

Geometry	Full upright original-mesh.stl
Envelope	Approximately 210.27 x 24.15 x 40.22 mm
Mesh	134,566 first-order tetrahedra, nominal 2.4 mm
Dense PLA	$E = 3.0 \text{ GPa}$; $\nu = 0.35$; $\rho = 1240 \text{ kg/m}^3$
Load	100 N downward on a finite top-midspan patch
Supports	Left 3D bearing; right depth and vertical roller constraints

2.1 Sparse shell/core representation

The dense shell volume is matched to two slicer perimeter widths ($0.42 + 0.45 \text{ mm}$). Each core uses the screening relationship $E_{\text{core}}/E_{\text{PLA}} = \text{relative density squared}$. A pattern-specific correction uses the G-code axial fourth-orientation moment normalized to cubic at the same density.

3. Results

Case	Tetrahedra	Mean disp. (mm)	Max. disp. (mm)	Stiffness (N/mm)	Mass (g)
Cubic 15%	134,566	0.39089	0.40766	255.83	40.93
Gyroid 15%	134,566	0.38626	0.40255	258.89	40.93
Honeycomb 15%	134,566	0.39057	0.40731	256.04	40.93
Cubic 25%	134,566	0.32648	0.33746	306.30	47.68
Gyroid 25%	134,566	0.31754	0.32790	314.92	47.68
Honeycomb 25%	134,566	0.32270	0.33341	309.89	47.68
Dense 100%	134,566	0.12432	0.12749	804.40	98.27

Table 1. Accepted fine-mesh results under the common 100 N load.

MBB 3DP FEM | Technical Validation Report

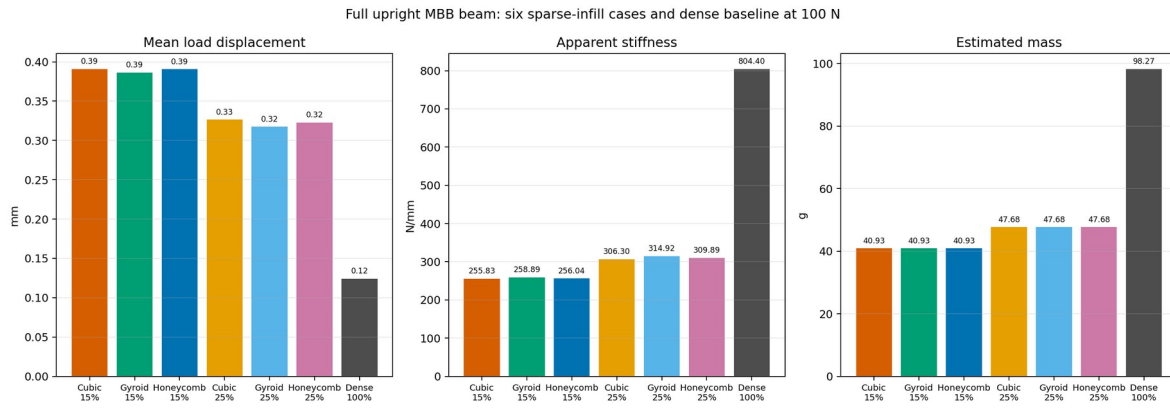


Figure 1. Full-upright displacement, apparent stiffness, and estimated mass comparison.

3.1 Engineering Comparison Matrix

Case	Infill	Mean disp. (mm)	Apparent stiffness (N/mm)	Stiffness vs. dense	Mass (g)	Specific stiffness (N/mm/g)	Rank
Cubic 15%	15%	0.39089	255.83	31.8%	40.93	6.25	7
Gyroid 15%	15%	0.38626	258.89	32.2%	40.93	6.32	5
Honeycomb 15%	15%	0.39057	256.04	31.8%	40.93	6.26	6
Cubic 25%	25%	0.32648	306.30	38.1%	47.68	6.42	4
Gyroid 25%	25%	0.31754	314.92	39.1%	47.68	6.61	2
Honeycomb 25%	25%	0.32270	309.89	38.5%	47.68	6.50	3
Dense 100%	100%	0.12432	804.40	100.0%	98.27	8.19	1

Table 2. Cross-case comparison under the common 100 N load. The blue row identifies the best sparse-case stiffness.

Dense stiffness reports each case as a percentage of the 100% solid baseline. Specific stiffness is apparent stiffness divided by estimated mass and is provided for comparative screening only. The sparse-core properties remain homogenized and require physical calibration.

4. Numerical Verification

Case	Reaction error	Energy error	Status
Cubic 15%	6.23e-09	1.08e-11	PASS
Gyroid 15%	5.52e-09	9.23e-12	PASS
Honeycomb 15%	6.20e-09	1.07e-11	PASS
Cubic 25%	8.78e-09	8.34e-12	PASS
Gyroid 25%	1.08e-08	7.24e-12	PASS
Honeycomb 25%	1.09e-08	7.34e-12	PASS
Dense 100%	1.63e-08	2.43e-11	PASS

Table 3. Reaction-equilibrium and strain-energy identity checks.

All three solvers converged, total vertical reactions reproduce the 100 N applied load, and energy-identity errors are below $2.5\text{e-}11$. No degenerate tetrahedra were reported.

4.1 Mesh sensitivity

Refining the sparse-case mesh from 3.0 to 2.4 mm changed mean displacement by 3.43-3.53% for the 15% cases and 2.15-2.29% for the 25% cases. The 15% results are close to, but slightly above, a provisional 3% convergence target.

5. Engineering Interpretation

Gyroid is the stiffest sparse screening pattern at both densities. Pattern-to-pattern mean-displacement spread is only 1.2% at 15% and 2.8% at 25%, so the ranking should be treated as provisional until calibrated.

Estimated mass is 40.93 g at 15% and 47.68 g at 25% for every pattern because the common shell and nominal density are held fixed. Pattern differences therefore arise only from the G-code orientation correction in this model.

USE OF RESULTS Use displacement and stiffness for comparative screening. Do not use the reported peak von Mises stress as a printed PLA failure criterion.

6. Limitations and Required Validation

Current limitation	Required next evidence
Homogenized pattern cores	Pattern-specific printed coupons or RVE elastic calibration
No layer-interface failure model	Upright tensile/shear interface characterization
Linear-elastic small strain	Physical three-point bending load-displacement test
Mesh-sensitive local stress	Local refinement and stress-convergence study
No print defects or porosity	Measured specimen mass, microscopy, and process record

7. Conclusion

Under the stated shell/core assumptions, gyroid provides the highest sparse-case stiffness, followed by honeycomb and cubic at both densities. The differences are modest, while the dense 100% beam remains substantially stiffer. Numerical balance checks pass for all accepted cases, supporting their use for comparative FEM screening.

The next defensible step is experimental calibration using full upright printed specimens with recorded slicer settings, measured mass, and three-point bending load-displacement data.

8. Reproducibility

Solver	src/fem/mbb_3d_pattern_shell_core.py
Dense baseline	src/fem/mbb_3d_dolfinx.py
G-code inspector	src/validation/inspect_infill_gcode_3mf.py
Comparison data	outputs/fem/all_patterns_full_upright_comparison/comparison.json
Detailed validation	outputs/fem/all_patterns_full_upright_comparison/validation.md

References

1. DOLFINx documentation. <https://docs.fenicsproject.org/dolfinx/main/python/>
2. Wu, J., Clausen, A., and Sigmund, O. (2017). Minimum compliance topology optimization of shell-infill composites for additive manufacturing. Computer Methods in Applied Mechanics and Engineering, 326, 358-375. <https://doi.org/10.1016/j.cma.2017.08.018>

Appendix A. Inspected .3MF Infill Toolpaths

Each figure is reconstructed directly from sparse-infill extrusion moves in Metadata/plate_1.gcode. The 25% archives are pattern-calibration evidence only; all FEM cases use the same full upright beam.

Cubic 15% - full upright archive

Sparse-infill toolpath verification
OG beam - 15% ID - CUBIC .gcode.3mf

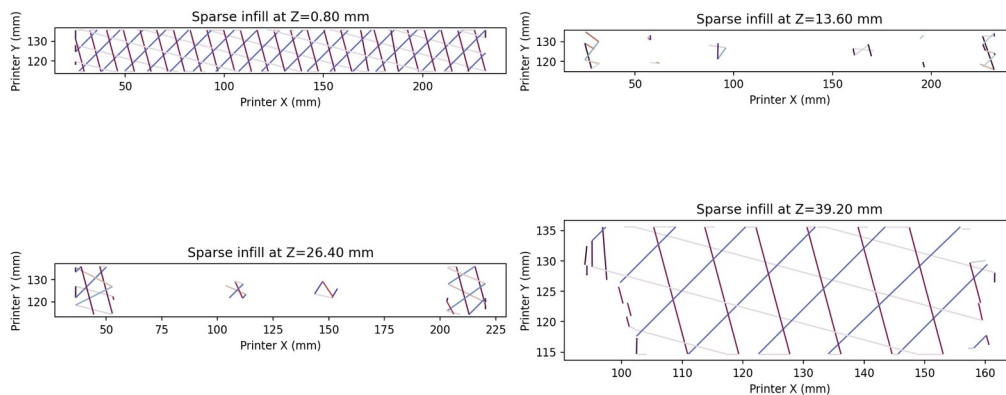


Figure A1. Inspected sparse-infill layers from the archived slicer .3mf file.

Cubic 25% - half-thickness calibration archive

Sparse-infill toolpath verification
CUBIC BEAM - 25% - HALF - YE .gcode.3mf

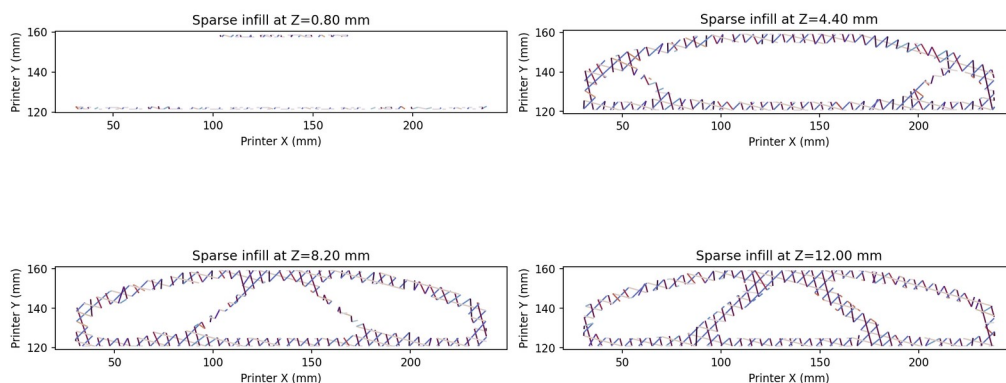


Figure A2. Inspected sparse-infill layers from the archived slicer .3mf file.

Gyroid 15% - full upright archive

Sparse-infill toolpath verification
OG beam - 15% ID - GYROID.gcode.3mf

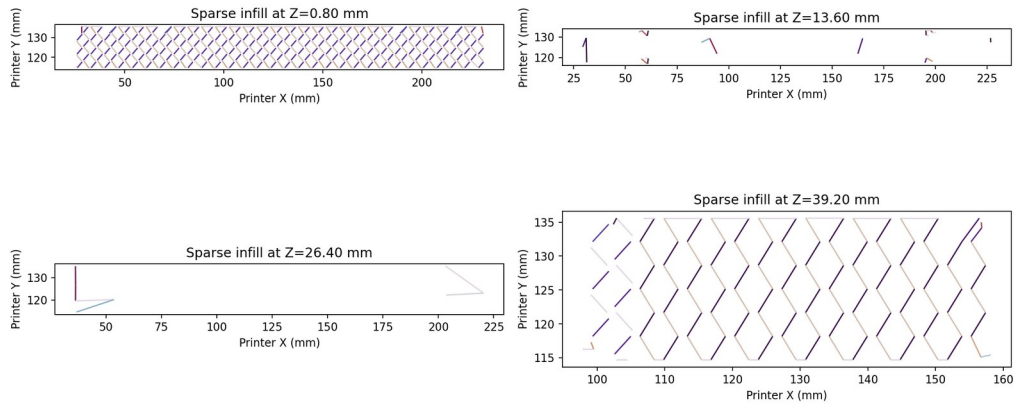


Figure A3. Inspected sparse-infill layers from the archived slicer .3mf file.

Gyroid 25% - half-thickness calibration archive

Sparse-infill toolpath verification
GYROID BEAM - 25% - HALF - YE gcode.3mf

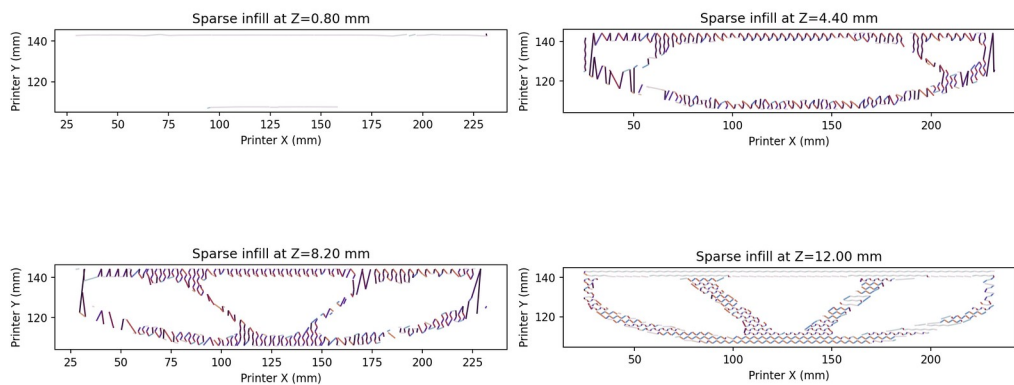


Figure A4. Inspected sparse-infill layers from the archived slicer .3mf file.

Honeycomb 15% - full upright archive

Sparse-infill toolpath verification
OG beam - 15% ID - HONEYCOMB - grid.gcode.3mf

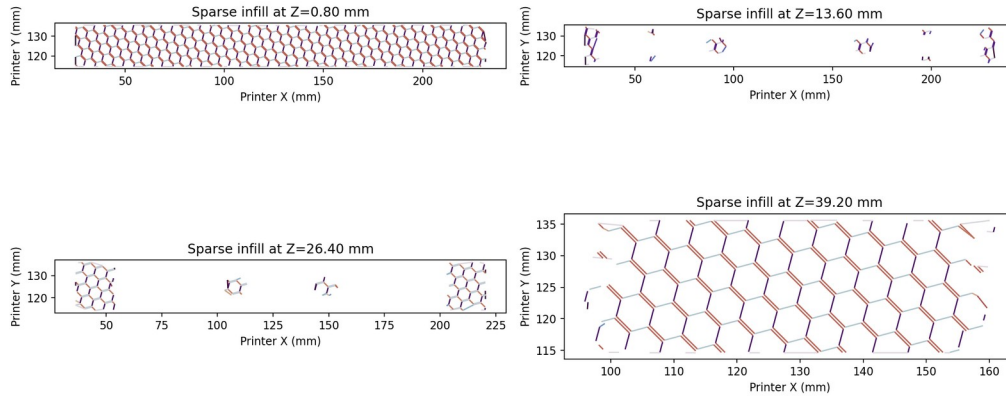


Figure A5. Inspected sparse-infill layers from the archived slicer .3mf file.

Honeycomb 25% - half-thickness calibration archive

Sparse-infill toolpath verification
HONEYCOMB BEAM - 25% - HALF .gcode.3mf

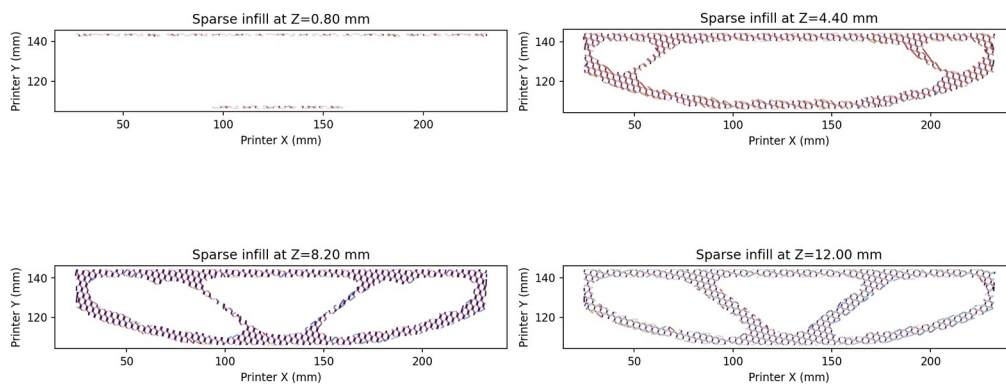


Figure A6. Inspected sparse-infill layers from the archived slicer .3mf file.

Appendix B. Three-Dimensional Meshes and FE Results

All six sparse material cases and the dense baseline use the same full-upright domain and 134,566-element tetrahedral discretization. The mesh view shows the exterior facets of that volume mesh; the result view shows displacement magnitude on the scaled deformed geometry.

Cubic 15%

Cubic 15% full upright
134,566 first-order tetrahedra | 44,378 exterior facets

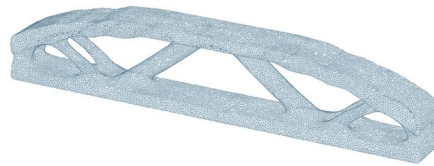


Figure B1. Cubic 15% 3D FE mesh: 134,566 first-order tetrahedra.

Full upright MBB cubic 15%
homogenized shell/core | deformation x15

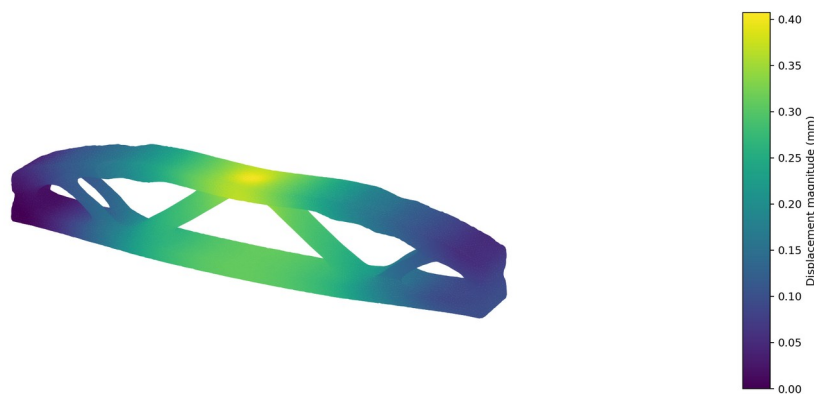


Figure B2. Cubic 15% FE displacement result: maximum displacement 0.4077 mm under the common 100 N load.

Cubic 25%

Cubic 25% full upright
134,566 first-order tetrahedra | 44,378 exterior facets

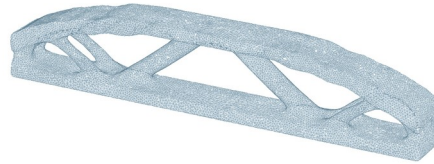


Figure B3. Cubic 25% 3D FE mesh: 134,566 first-order tetrahedra.

Full upright MBB cubic 25%
homogenized shell/core | deformation x15

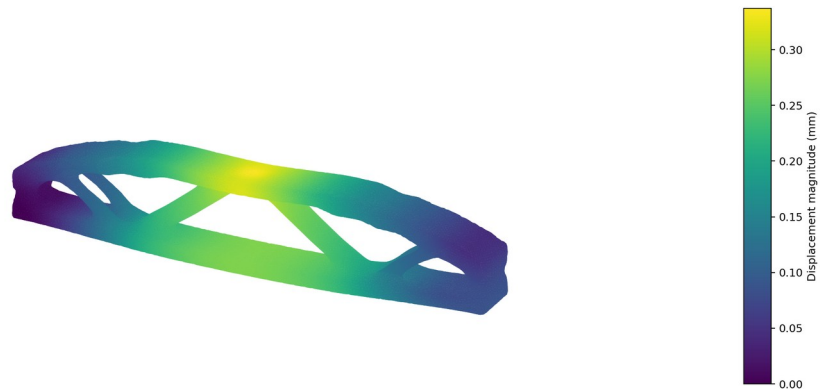


Figure B4. Cubic 25% FE displacement result: maximum displacement 0.3375 mm under the common 100 N load.

Gyroid 15%

Gyroid 15% full upright
134,566 first-order tetrahedra | 44,378 exterior facets

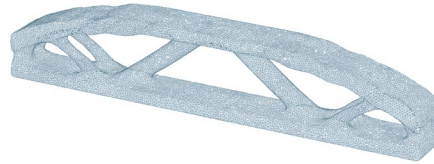


Figure B5. Gyroid 15% 3D FE mesh: 134,566 first-order tetrahedra.

Full upright MBB gyroid 15%
homogenized shell/core | deformation x15

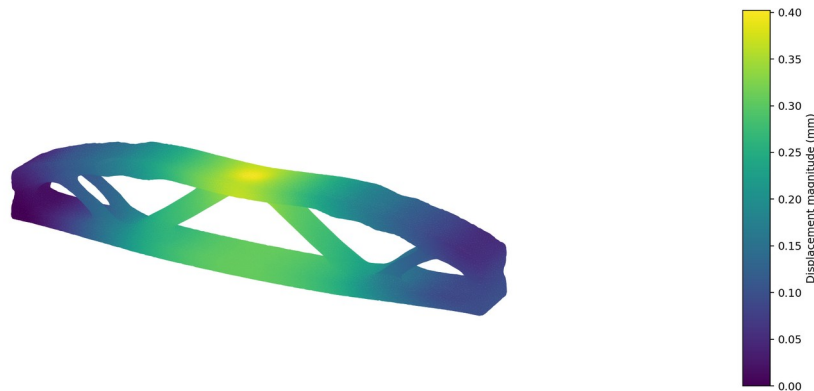


Figure B6. Gyroid 15% FE displacement result: maximum displacement 0.4025 mm under the common 100 N load.

Gyroid 25%

Gyroid 25% full upright
134,566 first-order tetrahedra | 44,378 exterior facets

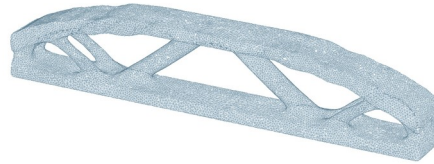


Figure B7. Gyroid 25% 3D FE mesh: 134,566 first-order tetrahedra.

Full upright MBB gyroid 25%
homogenized shell/core | deformation x15

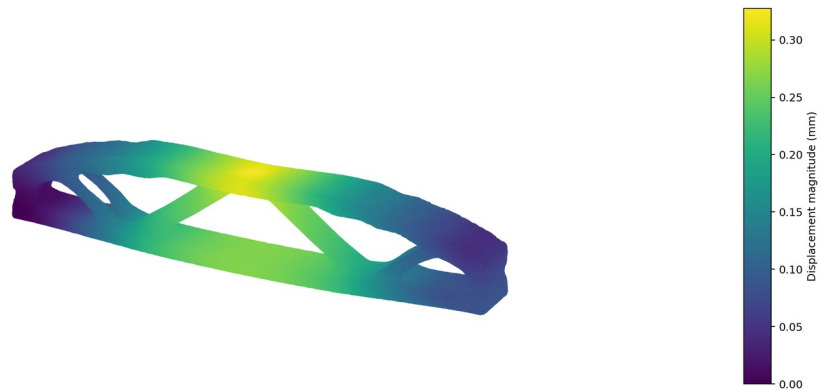


Figure B8. Gyroid 25% FE displacement result: maximum displacement 0.3279 mm under the common 100 N load.

Honeycomb 15%

Honeycomb 15% full upright
134,566 first-order tetrahedra | 44,378 exterior facets

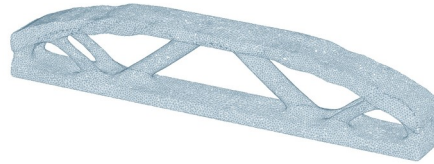


Figure B9. Honeycomb 15% 3D FE mesh: 134,566 first-order tetrahedra.

Full upright MBB honeycomb 15%
homogenized shell/core | deformation x15

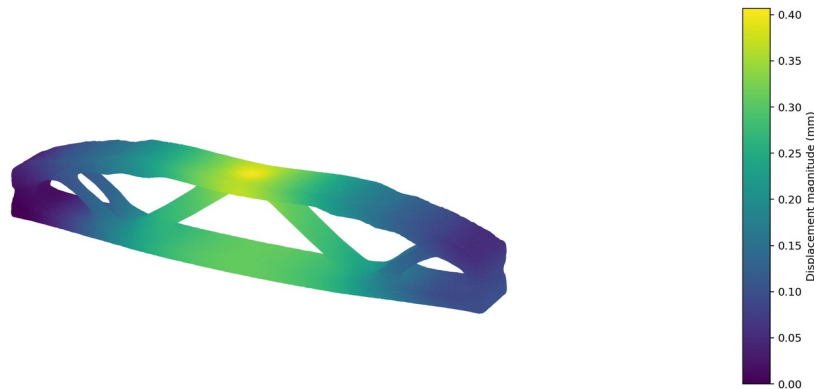


Figure B10. Honeycomb 15% FE displacement result: maximum displacement 0.4073 mm under the common 100 N load.

Honeycomb 25%

Honeycomb 25% full upright
134,566 first-order tetrahedra | 44,378 exterior facets

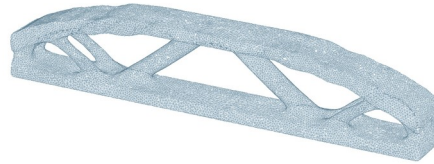


Figure B11. Honeycomb 25% 3D FE mesh: 134,566 first-order tetrahedra.

Full upright MBB honeycomb 25%
homogenized shell/core | deformation x15

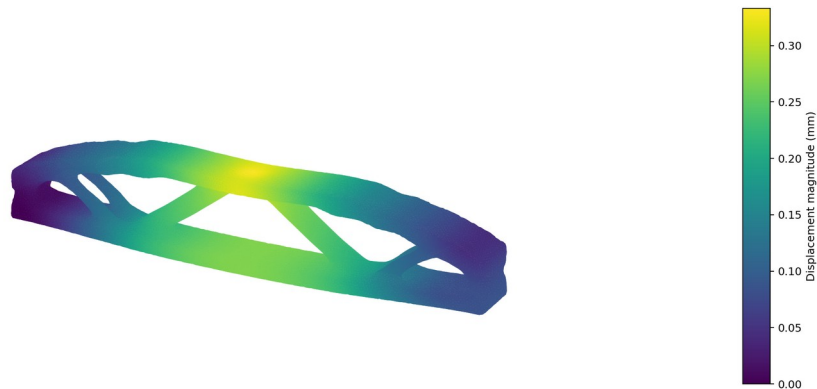


Figure B12. Honeycomb 25% FE displacement result: maximum displacement 0.3334 mm under the common 100 N load.

Dense 100% baseline

Dense 100% full upright baseline
134,566 first-order tetrahedra | 44,378 exterior facets

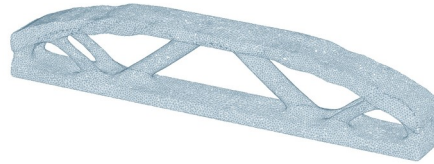


Figure B13. Dense 100% baseline 3D FE mesh: 134,566 first-order tetrahedra.

Dense MBB true 3D FEM
100% cubic-equivalent solid | deformation x75

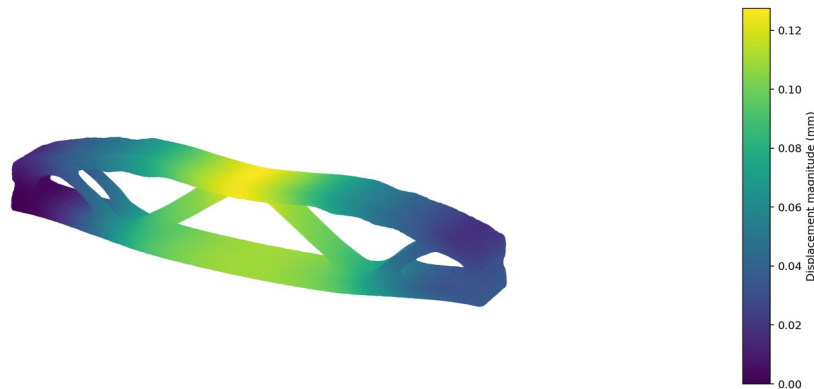


Figure B14. Dense 100% baseline FE displacement result: maximum displacement 0.1275 mm under the common 100 N load.